

## KYZYLORDA, Kazakhstan

WMO: 380620

Lat: 44.8247N

Lon: 65.5272E

Elev: 427

StdP: 14.47

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|------------|-------------|------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |            | 1%          |            |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-9.5 | (c)<br>-4.1 | (d)<br>-15.7                  | (e)<br>2.4 | (f)<br>-7.6 | (g)<br>-10.9 | (h)<br>3.1 | (i)<br>-3.2 | (j)<br>31.0           | (k)<br>8.4 | (l)<br>27.3 | (m)<br>7.0 | (n)<br>8.2            | (o)<br>40 | (p)<br>0.537 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |           |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|-----------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |           |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD      |
| (a)<br>7      | (b)<br>25.4            | (c)<br>102.4    | (d)<br>67.9 | (e)<br>99.2 | (f)<br>66.8 | (g)<br>96.5 | (h)<br>65.9 | (i)<br>70.9         | (j)<br>94.4 | (k)<br>69.0 | (l)<br>93.1 | (m)<br>67.4 | (n)<br>91.9 | (o)<br>9.2           | (p)<br>70 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.8                            | 86.7 | 81.6 | 60.1 | 78.9 | 80.2 | 57.5 | 71.6 | 78.9 | 35.0          | 93.8 | 33.4 | 93.2 | 32.1 | 91.9 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 24.3              | 21.0 | 17.9 | -14.0                      | 107.4 | 7.7                | 3.1 | -19.5                                              | 109.6 | -24.0      | 111.5 | -28.3      | 113.2 | -33.9      | 115.4 |
|                   |      |      | -13.8                      | 74.2  | 8.9                | 4.4 | -20.2                                              | 77.4  | -25.5      | 80.0  | -30.5      | 82.5  | -37.0      | 85.7  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 52.0   | 19.5  | 23.8  | 40.1  | 56.3  | 69.2  | 78.5  | 81.1  | 78.1  | 65.7  | 51.0  | 35.8  | 23.5  |  |  |
|                                                                        | DBStd       | 23.91  | 11.72 | 12.91 | 11.36 | 9.75  | 7.23  | 6.41  | 5.69  | 6.60  | 7.90  | 9.13  | 10.52 | 11.00 |  |  |
|                                                                        | HDD50       | 3429   | 945   | 734   | 339   | 50    | 1     | 0     | 0     | 0     | 2     | 99    | 438   | 821   |  |  |
|                                                                        | HDD65       | 6357   | 1410  | 1153  | 773   | 286   | 41    | 2     | 0     | 4     | 86    | 439   | 877   | 1286  |  |  |
|                                                                        | CDD50       | 4172   | 0     | 0     | 33    | 240   | 595   | 854   | 964   | 871   | 473   | 131   | 11    | 0     |  |  |
|                                                                        | CDD65       | 1623   | 0     | 0     | 1     | 26    | 170   | 406   | 499   | 409   | 106   | 6     | 0     | 0     |  |  |
|                                                                        | CDH74       | 22699  | 0     | 0     | 32    | 432   | 2256  | 5708  | 7127  | 5575  | 1435  | 134   | 0     | 0     |  |  |
|                                                                        | CDH80       | 12170  | 0     | 0     | 7     | 134   | 968   | 3183  | 4134  | 3121  | 595   | 28    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 7.6    | 7.6   | 7.4   | 8.6   | 8.8   | 8.7   | 7.4   | 7.1   | 7.8   | 7.2   | 6.5   | 6.6   | 7.3   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 5.8    | 0.6   | 0.5   | 0.7   | 0.7   | 0.7   | 0.3   | 0.2   | 0.1   | 0.1   | 0.5   | 0.6   | 0.7   |  |  |
|                                                                        | PrecMax     | 11.9   | 1.6   | 1.9   | 2.1   | 2.6   | 3.2   | 2.5   | 1.1   | 0.5   | 0.7   | 1.8   | 1.7   | 2.0   |  |  |
|                                                                        | PrecMin     | 2.8    | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 2.0    | 0.4   | 0.4   | 0.5   | 0.7   | 0.8   | 0.5   | 0.2   | 0.1   | 0.2   | 0.5   | 0.4   | 0.4   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 47.2  | 58.8  | 78.8  | 90.7  | 97.6  | 104.9 | 107.8 | 104.3 | 97.7  | 83.9  | 67.9  | 49.9  |  |  |
|                                                                        |             | MCWB   | 40.7  | 47.5  | 57.6  | 63.3  | 64.2  | 68.6  | 69.7  | 68.5  | 65.4  | 59.0  | 54.1  | 41.7  |  |  |
|                                                                        | 2%          | DB     | 40.5  | 49.0  | 69.5  | 83.9  | 92.9  | 100.8 | 103.6 | 100.5 | 91.4  | 77.4  | 60.4  | 44.4  |  |  |
|                                                                        |             | MCWB   | 36.7  | 42.5  | 53.3  | 59.6  | 63.1  | 66.9  | 68.4  | 67.4  | 62.6  | 55.9  | 50.2  | 39.7  |  |  |
|                                                                        | 5%          | DB     | 36.8  | 43.6  | 62.9  | 78.9  | 89.3  | 97.4  | 99.5  | 97.5  | 86.7  | 73.0  | 55.2  | 40.1  |  |  |
|                                                                        |             | MCWB   | 34.3  | 38.2  | 49.8  | 57.5  | 61.5  | 66.2  | 67.2  | 66.2  | 60.8  | 54.3  | 46.4  | 36.7  |  |  |
|                                                                        | 10%         | DB     | 33.7  | 39.4  | 57.5  | 74.4  | 85.7  | 94.4  | 96.3  | 94.5  | 82.3  | 67.9  | 50.2  | 36.6  |  |  |
|                                                                        |             | MCWB   | 31.6  | 35.1  | 47.3  | 55.6  | 60.7  | 65.3  | 66.2  | 65.1  | 59.1  | 51.7  | 43.7  | 34.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 41.6  | 49.6  | 58.1  | 65.2  | 69.8  | 73.8  | 74.0  | 72.6  | 66.5  | 60.1  | 56.1  | 44.0  |  |  |
|                                                                        |             | MCDB   | 45.9  | 57.9  | 75.1  | 83.1  | 87.6  | 96.0  | 96.3  | 96.1  | 93.7  | 80.4  | 64.4  | 46.6  |  |  |
|                                                                        | 2%          | WB     | 37.4  | 43.0  | 54.2  | 61.4  | 65.4  | 70.3  | 70.9  | 69.3  | 63.6  | 57.5  | 51.2  | 40.2  |  |  |
|                                                                        |             | MCDB   | 40.0  | 47.9  | 66.5  | 78.3  | 85.6  | 92.4  | 95.1  | 95.7  | 88.1  | 74.7  | 58.8  | 43.7  |  |  |
|                                                                        | 5%          | WB     | 34.4  | 38.9  | 50.9  | 59.0  | 63.4  | 68.2  | 69.1  | 67.6  | 61.6  | 54.9  | 47.1  | 37.1  |  |  |
|                                                                        |             | MCDB   | 36.2  | 43.3  | 61.8  | 76.6  | 84.7  | 91.4  | 93.5  | 93.1  | 84.6  | 70.7  | 53.6  | 39.8  |  |  |
|                                                                        | 10%         | WB     | 31.7  | 35.4  | 47.7  | 56.5  | 61.7  | 66.5  | 67.5  | 66.0  | 59.6  | 52.4  | 43.7  | 34.2  |  |  |
|                                                                        |             | MCDB   | 33.4  | 38.8  | 57.0  | 72.3  | 82.8  | 90.5  | 92.2  | 91.4  | 81.0  | 66.5  | 49.7  | 36.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.1  | 17.3  | 20.7  | 23.7  | 25.3  | 25.6  | 25.4  | 26.6  | 26.7  | 24.6  | 17.6  | 14.2  |  |  |
|                                                                        |             | MCDBR  | 15.0  | 20.8  | 29.5  | 29.9  | 30.2  | 29.8  | 30.4  | 30.6  | 32.3  | 31.1  | 24.8  | 17.3  |  |  |
|                                                                        | 5% WB       | MCWBR  | 11.7  | 14.6  | 16.7  | 14.5  | 12.5  | 11.6  | 11.5  | 12.3  | 14.3  | 17.1  | 15.6  | 13.2  |  |  |
|                                                                        |             | MCWBR  | 13.4  | 19.5  | 27.2  | 27.0  | 27.0  | 26.5  | 27.4  | 28.2  | 30.0  | 28.7  | 22.2  | 15.1  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.351 | 0.379 | 0.435 | 0.478 | 0.482 | 0.462 | 0.444 | 0.426 | 0.393 | 0.398 | 0.381 | 0.356 |  |  |
|                                                                        | taud        |        | 2.066 | 2.033 | 1.981 | 1.912 | 1.949 | 2.056 | 2.130 | 2.154 | 2.188 | 2.094 | 2.101 | 2.085 |  |  |
|                                                                        | Ebn at Noon |        | 226   | 245   | 248   | 248   | 252   | 258   | 261   | 261   | 260   | 235   | 213   | 209   |  |  |
|                                                                        | Edh at Noon |        | 33    | 41    | 49    | 58    | 57    | 52    | 48    | 45    | 40    | 38    | 32    | 29    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 447   | 817   | 1248  | 1724  | 2192  | 2373  | 2332  | 2104  | 1652  | 1056  | 597   | 394   |  |  |
|                                                                        | RadStd      |        | 89    | 99    | 95    | 130   | 124   | 111   | 101   | 71    | 55    | 63    | 58    | 59    |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |  | N/A   | N/A     | N/A    | +2.12   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) |  | N/A   | N/A     | N/A    | +1.53   | N/A   | -1.94 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page